# A UNIVERSALLY CLOSED MORPHISM OF ALGEBRAIC SPACES OVER A SCHEME IS QUASI-COMPACT

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof opens with the author's remark that it repeats the proof of the scheme case in Morphisms, Lemma morphisms-lemma-universally-closed-quasi-compact; that sentence is kept verbatim and the argument following it is written out in full, so nothing counted rests on the pointer.

*Editorial note.* The seven references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

*Editorial note.* Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, the reference/slogan/history comment environments; 14-18, the Xy-pic setup; 154-217, the theorem declarations and mathematical macros). The chapter file itself carries no preamble text.

*Allowed standard prerequisites (editorial).*

- Quasi-compactness of a morphism of algebraic spaces and its criterion local on the target. Lemma lemma-quasi-compact-local, used at proof lines 1696-1700 and 1734-1737. Reduces the problem to the case Y = Spec(B) and converts the failure of quasi-compactness into a contradiction.
- Permanence of quasi-compactness for the morphisms X_i -> Y. Lemma lemma-quasi-compact-permanence, used at proof lines 1707-1708. Supplies the quasi-compactness of the pieces of the chosen open covering.
- Reduced closed subspaces of an algebraic space cut out by a closed set of points. Properties of Spaces, Lemma spaces-properties-lemma-reduced-closed-subspace, used at proof lines 1710-1714. Produces the subspace Z whose image is shown not to be closed.
- Points of fibre products of algebraic spaces and their field-valued representatives. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, used at proof lines 1763-1765. Lifts the pair (x, t') to a point z of X x_Y T lying in |Z|.
- Base change of open immersions and of the diagram T x_Y X -> Y. Spaces, Lemmas spaces-lemma-base-change-immersions and spaces-lemma-composition-immersions, used at proof lines 1715-1718. Identifies T_i x_Y X_i as an open subspace so that Z is cut out by its complement.

*Source reference.* Original lemma 03KG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4676–4676. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

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*Date*: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 03KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4688–4688. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 04XU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6407–6407. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02YW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3891–3891. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02YV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3890–3890. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 03H4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4553–4553. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 03IQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4612–4612. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

**Lemma 0.2.** *Let $S$ be a scheme. A universally closed morphism of algebraic spaces over $S$ is quasi-compact.*

*Proof.* This proof is a repeat of the proof in the case of schemes, see Morphisms, Lemma 04XU. Let $f : X \to Y$ be a morphism of algebraic spaces over $S$. Assume that $f$ is not quasi-compact. Our goal is to show that $f$ is not universally closed. By Lemma 03KG there exists an affine scheme $Z$ and a morphism $Z \to Y$ such that $Z \times_Y X \to Z$ is not quasi-compact. To achieve our goal it suffices to show that $Z \times_Y X \to Z$ is not universally closed, hence we may assume that $Y = \mathrm{Spec}(B)$ for some ring $B$.

Write $X = \bigcup_{i \in I} X_i$ where the $X_i$ are quasi-compact open subspaces of $X$. For example, choose a surjective étale morphism $U \to X$ where $U$ is a scheme, choose an affine open covering $U = \bigcup U_i$ and let $X_i \subset X$ be the image of $U_i$. We will use later that the morphisms $X_i \to Y$ are quasi-compact, see Lemma 03KS. Let $T = \mathrm{Spec}(B[a_i; i \in I])$. Let $T_i = D(a_i) \subset T$. Let $Z \subset T \times_Y X$ be the reduced closed subspace whose underlying closed set of points is $|T \times_Y Z| \setminus \bigcup_{i \in I} |T_i \times_Y X_i|$, see Properties of Spaces, Lemma 03IQ. (Note that $T_i \times_Y X_i$ is an open subspace of $T \times_Y X$ as $T_i \to T$ and $X_i \to X$ are open immersions, see Spaces, Lemmas 02YW and 02YV.) Here is a diagram

$$\begin{array}{ccccc} Z & \longrightarrow & T \times_Y X & \xrightarrow[q]{} & X \\ & \searrow & \downarrow{\scriptstyle f_T} & & \downarrow{\scriptstyle f} \\ & & T & \xrightarrow{p} & Y \end{array}$$

It suffices to prove that the image $f_T(|Z|)$ is not closed in $|T|$.

We claim there exists a point $y \in Y$ such that there is no affine open neighborhood $V$ of $y$ in $Y$ such that $X_V$ is quasi-compact. If not then we can cover $Y$ with finitely many such $V$ and for each $V$ the morphism $Y_V \to V$ is quasi-compact by Lemma 03KS and then Lemma 03KG implies $f$ quasi-compact, a contradiction. Fix a $y \in Y$ as in the claim.

Let $t \in T$ be the point lying over $y$ with $\kappa(t) = \kappa(y)$ such that $a_i = 1$ in $\kappa(t)$ for all $i$. Suppose $z \in |Z|$ with $f_T(z) = t$. Then $q(t) \in X_i$ for some $i$. Hence $f_T(z) \notin T_i$ by construction of $Z$, which contradicts the fact that $t \in T_i$ by construction. Hence we see that $t \in |T| \setminus f_T(|Z|)$.

Assume $f_T(|Z|)$ is closed in $|T|$. Then there exists an element $g \in B[a_i; i \in I]$ with $f_T(|Z|) \subset V(g)$ but $t \notin V(g)$. Hence the image of $g$ in $\kappa(t)$ is nonzero. In particular some coefficient of $g$ has nonzero image in $\kappa(y)$. Hence this coefficient is invertible on some affine open neighborhood $V$ of $y$. Let $J$ be the finite set of $j \in I$ such that the variable $a_j$ appears in $g$. Since $X_V$ is not quasi-compact and each $X_{i,V}$ is quasi-compact, we may choose a point $x \in |X_V| \setminus \bigcup_{j \in J} |X_{j,V}|$. In other words, $x \in |X| \setminus \bigcup_{j \in J} |X_j|$ and $x$ lies above some $v \in V$. Since $g$ has a coefficient that is invertible on $V$, we can find a point $t' \in T$ lying above $v$ such that $t' \notin V(g)$ and $t' \in V(a_i)$ for all $i \notin J$. This is true because $V(a_i; i \in I \setminus J) = \mathrm{Spec}(B[a_j; j \in J])$ and the set of points of this scheme lying over $v$ is bijective with $\mathrm{Spec}(\kappa(v)[a_j; j \in J])$ and $g$ restricts to a nonzero element of this polynomial ring by construction. In other words $t' \notin T_i$ for each $i \notin J$. By Properties of Spaces, Lemma 03H4 we can find a point $z$ of $X \times_Y T$ mapping to $x \in X$ and to $t' \in T$. Since $x \notin |X_j|$ for $j \in J$ and $t' \notin T_i$ for $i \in I \setminus J$ we see that $z \in |Z|$. On the other hand $f_T(z) = t' \notin V(g)$ which contradicts $f_T(Z) \subset V(g)$. Thus the assumption "$f_T(|Z|)$ closed" is wrong and we conclude indeed that $f_T$ is not closed as desired. $\square$